Exploring the Impact of AI-Generated Images on Political News Perception and Understanding

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ABSTRACT

In political news articles, images play an important role in conveying news content and attracting readers' attention. As image generation technology has been developed and its use has increased, this study investigated the effect of image generation of political news articles on readers' perception and response to the articles. We first examined the primary elements that characterize political news articles and used the latest text-to-image model with these elements to generate images appropriate for liberal/conservative media articles. The results of the user study with 102 participants showed that the generated images reflected the content of the articles better than the original images and facilitated the understanding of the articles. In particular, conservative participants preferred generated images in factual reports, and liberal participants preferred generated images in biased reports, suggesting a careful consideration of the use of generated images in information delivery. Our study opens up new possibilities for the use of AI in journalism, where providing fair and clear information to readers and reducing political polarization is important.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

Image generation, Text-to-image AI, Political news, Journalism

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1 INTRODUCTION

Images often serve as the first point of contact between the public and the news, due to their ability to capture visual interest and efficiently convey information [8]. As visual information plays a crucial role in shaping the public's perception of news content, people often form their initial understanding of an article through the headlines and images they first encounter [45, 47].

In political articles, the importance of images is even more pronounced. Images can have the ability to implicitly convey a political context that goes beyond the mere dissemination of information. However, the use of inappropriate images can mislead or bias the reader's perception [20, 26]. Therefore, given the potential for political bias stemming from images [27], it is crucial to use images that can clearly and fairly represent the political context.

Recent advances in text-to-image artificial intelligence technology, which combines natural language processing and computer vision, have enabled the real-time generation of diverse visual artwork based on user text input. With the popularization of image generation services such as MidJourney and DALL-E, the use of generated images in news articles has also increased. However, these services typically generate images based on generic prompts that may not adequately reflect the detailed and nuanced content of news articles. Given the role of images in political articles, generating appropriate images for news articles is an important aspect of responsible journalism. It is thus crucial to recognize the powerful influence of images, understand the factors that contribute to image generation [6, 36], and focus on providing readers with accurate information and insight.

The objective of this study is to investigate user perceptions of political news image generation. To this end, we propose a method of image generation based on the identification of key elements that are directly or indirectly present in political news articles. These elements include article tone, key keywords, linguistic features, and topics according to theoretical foundations [9, 12, 31, 40] . We then investigate the perception of the generated images to discuss directions for reliable image generation. For this purpose, we crawled political news articles from prominent liberal and conservative media outlets in the United States. To ensure a diverse range of

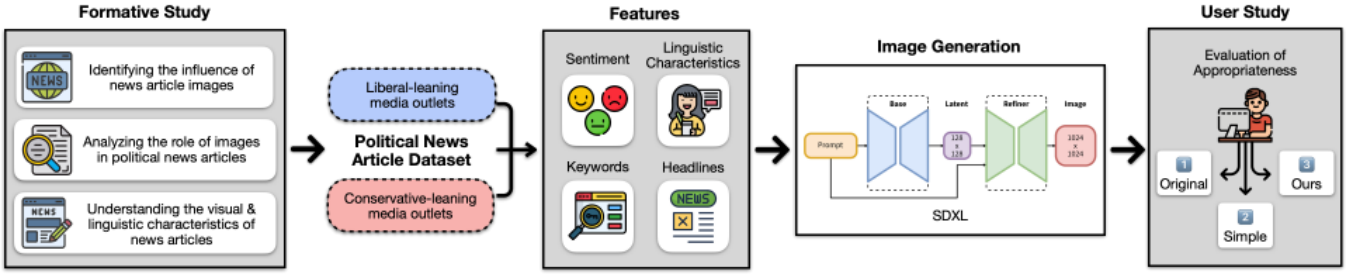


Figure 1: An overview of our research procedure. The study begins with formative research to analyze the role of images in political news articles and to understand their visual and linguistic characteristics. Next, we collect a dataset of political news articles from both liberal and conservative media outlets. From this collected data, we extract key features of the articles and generate prompts based on these features to create images using the Stable Diffusion XL (SDXL) model. Finally, to evaluate the appropriateness of the generated images, we conduct a user study comparing three types of images: (1) the original news image *(Original)*, (2) an image generated using the article content as a prompt *(Simple)*, and (3) an image generated using our method *(Ours)*.

political news articles, we considered two distinct categories: "biased reporting" (articles perceived as favoring one side of an issue) and "factual reporting" (reporting based on objective facts). We then quantified the essential components of the articles to create prompts and used the Stable Diffusion XL model to generate images. A formative study was conducted to select the images that most accurately reflected the article content. Using the selected images, we conducted a user study with 102 participants to evaluate the comprehension and effectiveness of the images.

As a result, the study participants responded that the images generated by our method better reflected the content of the news articles and facilitated comprehension than existing images of the news articles. In particular, participants with conservative leanings exhibited a higher preference for the generated images in the context of factual reporting. Conversely, for biased reporting, participants with liberal leanings showed a higher preference for the generated images. This result highlights the importance of image generation according to the type of reporting and the characteristics of the readers.

The contributions of this study are as follows:

- We explore a method that can contribute to the generation of news images that well reflect political news articles.
- We propose a new approach to public discourse by analyzing of the impact of generated images on readers' perception.
- We provide a new perspective on the use of AI in journalism and discuss effective methods for visual communication.

2 EXPERIMENT

2.1 Dataset

Among the major media outlets in the United States, we selected conservative and liberal news outlets for crawling political section news articles. To select media outlets, we referred to previous studies [10, 19]. As a result, we chose Breitbart (breitbart.com) and The Washington Post (washingtonpost.com) for the conservative and liberal sites, respectively. We crawled the political category articles from these two media outlets, collecting the publication date and time, title, article content, comments, article images, and article links. Basic preprocessing (e.g., remove special characters, change uppercase to lowercase) was then performed.

2.2 Key Element Extraction

2.2.1 Sentiment analysis. According to previous studies, there is a close relationship between sentiment, the color and composition of images [1, 9]. Therefore, the overall tone of an image significantly influences readers' perceptions and reactions, which may vary based on political orientation [15]. Identifying the tone of an article through sentiment analysis can serve as an indicator of the implicit tone conveyed by the image [37]. Additionally, from a political perspective, the political tone of an article can be reflected in the image, further increasing its importance [29]. We used a RoBERTa-based model trained on Kaggle's NewsMTSC dataset [17] to classify the sentiment into three categories: negative, neutral, and positive. Since the input data was limited to 512 tokens, in order to reflect the news article as much as possible, we summarized the news articles by fine-tuning Google's Pegasus-multi news model trained on the BBC News Summary dataset [46].

2.2.2 TF-IDF. Extracting keywords from political news articles plays a crucial role in clearly communicating complex political issues to readers [22, 24]. Political issues often involve complicated linguistic expressions related to various topics, and identifying the most important words or phrases helps to effectively communicate the core message of the news. Term Frequency-Inverse Document Frequency (TF-IDF) is an analytical method that considers both the frequency of a word and its rarity across a set of documents. By applying the results of this analysis to image generation, the essential and distinctive content of the article can be reflected in the image, which helps to visually emphasize the main points of the news [31].

2.2.3 Political linguistics. Each political orientation tends to use specific language and terms to express its values and beliefs [11, 39]. By applying these linguistic features to image generation, the political message of the article can be conveyed more clearly and accurately, helping readers better understand the political context of the article [43]. Therefore, it is important to consider the political

Table 1: This table summarizes the elements used for sentiment analysis, keyword extraction, linguistic characteristics, and headline analysis of news articles.

Factor	Purpose	Features	Method			Paper
Sentiment	Determine the emotional tone of the news article	Influence the mood and tone of the generated image	- Negative sentiment - Positive sentiment - Neutral sentiment			Dovdon et al. [9] Amencherla et al. [1]
Keywords	Extract keywords from the news article	Visualizes keywords in the generated image	Using keywords with a score of 0.15 or higher			Patel et al. [31]
Linguistic Characteristics	Reflect the political orientation of the news article	Reflect specific political messages in the generated image	Linguistic features of Liberals/Conservatives	Liberals	- Language Uniqueness - Emotional Empathy - Socio-Political Responsiveness - Fairness and Openness	Sterling et al. [40] Sylwester et al. [41] Gangula et al. [12] Rozado et al. [35] Groseclose et al. [14] Brugman et al. [4] Romer et al. [34]
				Conservatives	- Group Cohesion and Order - Religious and Traditional Values - Resistance and Certainty	
				Liberal Media	- Openness, Diversity and Fairness - Creativity and Wit	
				Conservative Media	- Religious and Traditional Values - Tradition, Order and Group Emphasis	
Headlines	Reflect the main themes of the news article	Reflect the core content in the generated image	News Media Headlines			Gangula et al. [12] Thomas et al. [42] Guo et al. [16]

orientation of the media outlet and its corresponding linguistic characteristics. We analyzed the differences in political bias in news language based on prior research [4, 12, 14, 34, 35] and systematically categorized these differences to apply them in the image generation process.

Based on the categorized information, we extracted relevant words from the news articles using Large Language Models (LLMs), ChatGPT. To select words most closely related to the content of the article, we excluded words that were not included in the top 100 words from the results of the TF-IDF analysis. This process allowed us to select keywords more closely related to the content of the news article, thereby increasing the accuracy of the analysis.

2.2.4 Headlines. The headline of a news article succinctly summarizes the article's topic, grabs the reader's attention, and encourages clicks. Journalists carefully craft headlines to best represent the content of the article [23, 38]. In political news, headlines tend to reflect the political orientation and stance of the article [12]. Liberal or conservative media outlets often reveal their political perspective through their headlines, sometimes with the use of specific language, even when covering the same event, thus influencing readers' perceptions and attitudes [16, 42]. In this sense, the linguistic choices and expressions in the headline play a crucial role in helping readers understand the topic of the news and the political position of the media outlet.

For these reasons, we used the headline in image generation to reflect the core content and tone of the article.

2.3 Image Generation

We used the Stable Diffusion XL (SDXL) model [32] for image generation. The SDXL model is suitable for generating images related to specific individuals or political events, making it ideal for generating images of political news articles. To generate images that match the news articles, we used prompts that included sentiment analysis results, TF-IDF extracted keywords, linguistic features by political orientation, and article headlines. To clearly convey the article content, we used a cartoon style, which helps to attract readers' attention and avoid offensive depictions [2, 5, 33]. Additionally, we included a negative prompt to prevent the inclusion of text for more intuitive images. Detailed examples of the prompts can be seen in Figure 2.

Prompt	Based on the article title 'What does a target letter mean for Trump? The latest on Jan. 6 probe', this image conveys a negative emotion. It visually represents key words from TF-IDF analysis such as 'charges', 'crimes', 'grand', 'indicted', 'jury', 'mcquade'. In the center of the image, there are piled legal documents, evidence items, and symbols of a courtroom, symbolizing phrases associated with progressive journalism such as 'another investigation', 'efforts to overturn', 'culminated in the attack', 'a lot we still don't know', 'exactly what it sounds like', 'declaration of rights', 'invitation to the recipient', 'heading in the direction', 'charges are coming', 'zeroed in on three big things', 'very serious crime', 'maximum of 20 years', 'a degree of independence', 'a grand jury must sign off', 'jury of your peers', 'in uncharted territory'. The image is also filled with elements symbolizing the legal proceedings related to President Trump, and their impact on American politics and society.
Negative Prompt	Please do not include any text, letters, or characters from any alphabet in the image. Avoid the use of words or numbers in any form or language.

Figure 2: An example of a prompt used in image generation. The red highlighted text indicates the headline of the article, the yellow highlighted text represents the emotion conveyed by the article, the green highlighted text lists the keywords extracted from the TF-IDF analysis, and the blue highlighted text shows the words included based on the writing characteristics of the news outlet's political orientation.

2.3.1 Image selection for user study. The guidance scale is a hyperparameter in text-to-image models that determines how closely the generated image should be "guided" by the text input. Higher values produce images that closely follow the prompt, while lower

Table 2: Categorization of the articles used in the user study by political orientation (liberal and conservative) and type of reporting (biased and factual).

		Biased reporting	Factual reporting
Trump	Liberal	What does a target letter mean for Trump? The latest on Jan. 6 probe	Trump attacks 'liberal Jews' amid celebration of Rosh Hashanah
	Conservative	Absolute Bullsh*t': GOP Lawmakers React as Trump Targeted in January 6 Probe	Trump Posts Rosh Hashanah Message from Conservative Jews Urging Liberal Jews to Repent
Biden	Liberal	Biden blasts Florida senators as he visits the state ahead of midterms	Biden interviewed about classified documents found at his office, home
	Conservative	Mr. Helpful: Joe Biden Bound for Florida to 'Boost' Democrat Chances	James Comer: Joe Biden Potentially Kept Sensitive Docs About Family Business Involvement with Foreign Countries

values produce more creative images [25, 30]. Therefore, lower values are used for tasks that require creativity, and higher values are used for images in educational materials or news articles where objectivity and accuracy are important [30]. We started with the default setting of 7.5 and increased it by 0.5, generating six images with different settings (see examples in Figure 3). To select the best representative images used in our main study, We conducted a user test with five participants. We presented them with the news article and the six images by the article, asking them to choose the most appropriate image.

2.4 User Study

We conducted a user study to evaluate user perception of the generated images. We targeted people in the United States and collected responses through Amazon Mechanical Turk from a total of 102 participants: 41 with liberal leanings, 41 with conservative leanings, and 20 with moderate leanings.

For the user study, we selected articles on the same topic about Donald Trump and Joe Biden from both conservative and liberal news outlets. Three researchers of this paper evaluated the articles and selected four articles for each category of (1) biased reporting (instances where political or personal views influence the coverage [3, 18]) and of (2) factual reporting (thoroughly verified facts with as little bias as possible [7]). Two articles described Donald Trump and the other two described Joe Biden. This allowed us to observe how participants responded to the image of each article and to analyze the relationship between participants' political orientation and image perception. The list of articles used in the study is summarized in Table 2.

We provided participants with the headline, body text, and three different images: (1) the original image used in the news article (*Original*), (2) an image generated using the entire article content as a prompt (*Simple*), and (3) an image generated using our method (*Ours*), which is selected in *Section* 2.3.1. The order of the images for each article was presented randomly. Participants were then asked to rate which of the three images best reflected the content of the article and helped them understand it. This process helps to verify the quality of the images generated by our method and to determine how much the generated images contribute to the understanding of the article content.

3 RESULTS

3.1 Image Generation

Figure 3 illustrates examples of images generated based on article content that varied according to the political orientation of news outlets. Conservative-(a) and Liberal-(a) were generated based on articles about President Joe Biden's activities related to the midterm elections. The conservative outlet focused on President Biden's visit to Florida and interactions with Democratic candidates, while the liberal outlet emphasized his speech and strategy. As a result, each image reflects a different political perspective, with one emphasizing a sunny background and the other an impassioned speech scene, illustrating differences in content based on political orientation. Additionally, Conservative-(b) and Liberal-(b) were generated based on articles about Trump's Rosh Hashanah post. The conservative outlet emphasized Trump's leadership and appeal, while the liberal outlet critically portrayed Jewish leaders' criticism and Trump's actions. As a result, the image generated based on the content of the liberal outlet appeared more confrontational than the one based on the conservative outlet's content.

These results show how conservative and liberal media outlets can interpret the same event differently, and highlight the impact of these content differences on image generation.

3.2 Evaluation of Appropriateness

The user study results showed that *Ours* (feature selection based on theoretical insights) better reflected the content of the article and supported comprehension more effectively than *Original* and *Simple*. Conversely, *Original* was preferred over *Simple*, possibly because participants found actual images more trustworthy and familiar. This is likely because *Original* emphasizes factuality and realism, which provides more credibility to participants. Nevertheless, the higher preference for *Ours* over *Original* indicates that our images also incorporated these characteristics of *Original*, confirming the effectiveness of our approach. Therefore, the high preference for *Ours* can be attributed not only to its creative and symbolic expressiveness, but also to its ability to connect the article content with the participants' political orientations.

To statistically analyze the differences in image perception by political orientation, we conducted a chi-square test (independent variable: political orientation *liberal, conservative*; dependent variable: image type *Original, Simple, Ours*). The results showed statistically significant differences in "Biased reporting" and "Factual reporting", with $\chi^2(2, N = 636) = 7.66$, $p = 0.022$ and $\chi^2(2, N = 636) = 9.10$, $p = 0.011$, respectively. This indicates that political orientation influences how individuals perceive media bias, particularly in their evaluations of the fairness and factuality of news articles. Liberal participants had a higher preference for *Ours* in biased reporting, while conservative participants preferred *Ours* in factual reporting.

This shows that liberals and conservatives may perceive the same article differently, which is also reflected in the type of images they prefer. Therefore, relevant stakeholders (e.g., media producers, reporters) should take these differences in political orientation into account when creating a representative image for the news article.

4 DISCUSSION

4.1 Study Implications: News Consumption Behavior and Media Strategy

Political orientation can lead to different preferences for factual reporting and biased reporting. These differences may stem from the values and information processing tendencies of each political

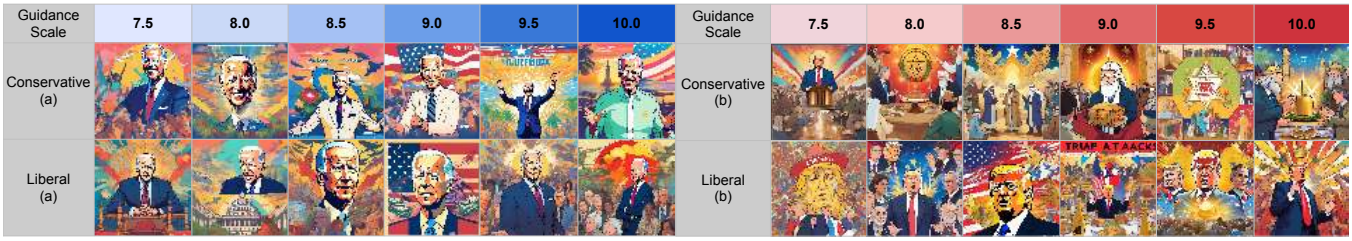


Figure 3: An example of the images generated based on news articles with different political orientations. Conservative-(a) and Liberal-(a) were created based on articles about President Joe Biden's midterm election activities. Conservative media focused on President Biden's visit to Florida and his interactions with Democratic candidates, while liberal media emphasized his speeches and strategies. Additionally, Conservative-(b) and Liberal-(b) were generated based on articles about Trump's Rosh Hashanah-related posts. Conservative media emphasized Trump's leadership and appeals, whereas liberal media critically portrayed the criticisms from Jewish leaders and Trump's actions.

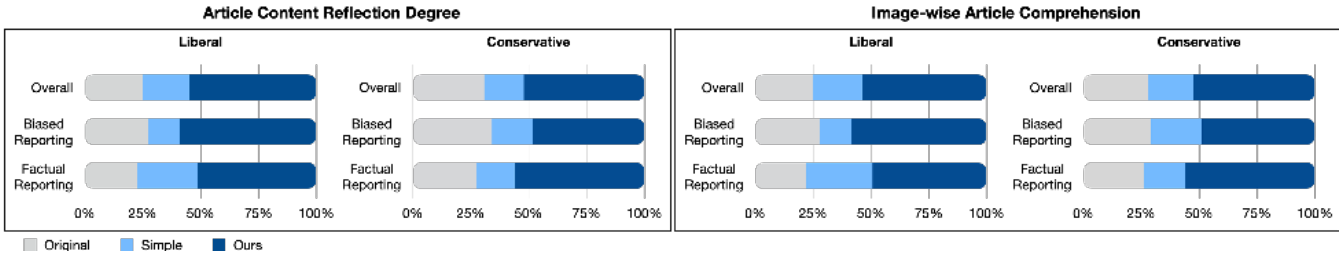


Figure 4: The results of the user study demonstrated that the images generated by our method (*Ours*) better reflect the content of the article and support comprehension more effectively than the *Original*. In particular, the *Ours* received higher ratings compared to the *Simple* and the *Original*.

group. In general, liberals value progressive change and diversity in society [28]. Because of these values, they tend to find news sources that align with their beliefs and validate their perspectives. They also tend to be more receptive to complex and multifaceted information than conservatives, thus they are more likely to prefer multi-view coverage [44]. On the other hand, conservatives value traditional values and order [13, 21]. Based on these values, they tend to trust objective and factual information and to prefer clear and specific information. These differences lead to different image preferences for each reporting type according to each group's political orientation.

Therefore, especially in the context of political news, it would be important for relevant stakeholders (e.g., media producers, reporters) to consider these differences in political orientation when creating a representative image for news content. It would also be important for them to take various approaches in order to choose the right direction for the fairness and reliability of information provision. To this end, it is necessary to understand the differences in media perception according to political orientation and to develop an effective journalism strategy based on it.

4.2 Future Directions: Enhancing News Consumption through Image Perception

This study evaluated the generation of appropriate images for political news articles and confirmed that our proposed method produced the most appropriate images from the readers' perspective. It should be noted that one of the limitations is that participants only experienced articles with generated images once. Future research aims to develop a web interface that allows participants to engage with generated images over time to gain a deeper understanding of the impact of images on perceptions and attitudes.

In addition, we plan to analyze users' aversion to images generated for articles with different political orientations. There is also a possibility of generating and using images for malicious purposes (e.g., promoting hostility and malice against others). This will help us explore ways to promote diversity in information consumption and mitigate possible negative effects (e.g., the echo chamber effect).

Our study findings highlight the influence of visual composition in news articles on readers' emotions and attitudes by discussing how the presentation of news affects public opinion and attitudes, and its applicability in the fields of journalism and social communication. Since our research objective is to improve the objectivity and fairness of news by increasing readers' access to various political perspectives, given the importance and ease of generating news images, developing and verifying a clear guideline for image generation in a news context, including journalistic and psychological perspectives, will be the next research steps.

5 CONCLUSION

In this paper, we developed an image generation method based on theoretical perspectives for political articles by analyzing political news from liberal and conservative media outlets in the United States. As a result of the user study, the images generated by our method *(Ours)* better reflected the content of the articles and enhanced understanding compared to *Simple* and *Original*. This indicates that AI-generated images can serve as a main article

representatives compared to traditional news images in delivering accurate and engaging content. We also confirmed statistically significant results in image preference by political orientation according to biased reporting and factual reporting, highlighting the importance of image generation according to the type of reporting and audience. In conclusion, *Ours* can significantly enhance readers' comprehension and engagement in political news articles, which (with additional careful verification) can be seen as a way to improve the clarity and impact of news reporting on politically sensitive topics.

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